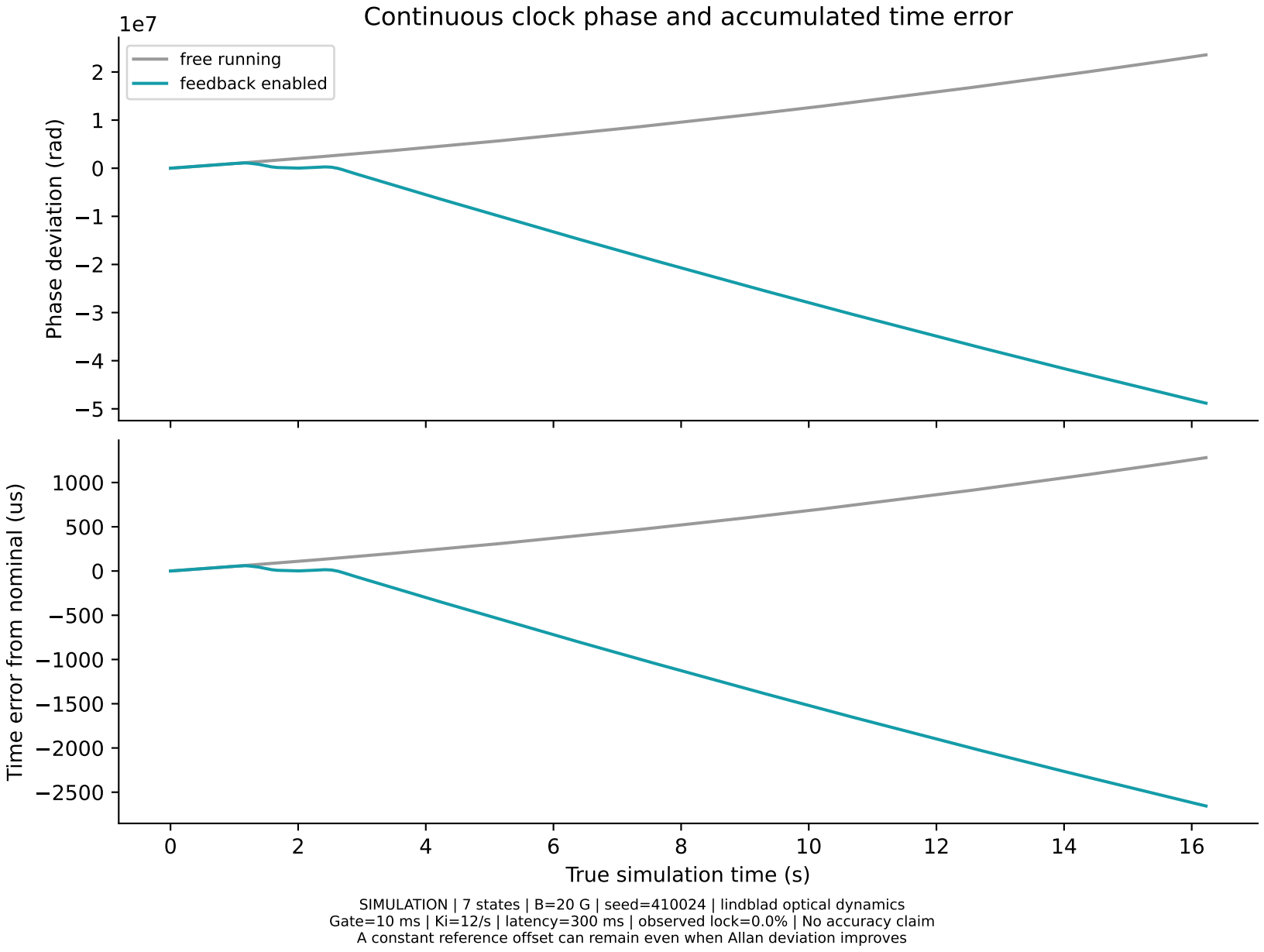
Continuous clock phase and accumulated time error
1e7
free running
feedback enabled
Phase deviation (rad)
2
1
0
−1
−2
−3
−4
−5
Time error from nominal (us)
1000
500
0
−500
−1000
−1500
−2000
−2500
0
2
4
6
8
10
12
14
16
True simulation time (s)
SIMULATION | 7 states | B=20 G | seed=410024 | lindblad optical dynamics
Gate=10 ms | Ki=12/s | latency=300 ms | observed lock=0.0% | No accuracy claim
A constant reference offset can remain even when Allan deviation improves